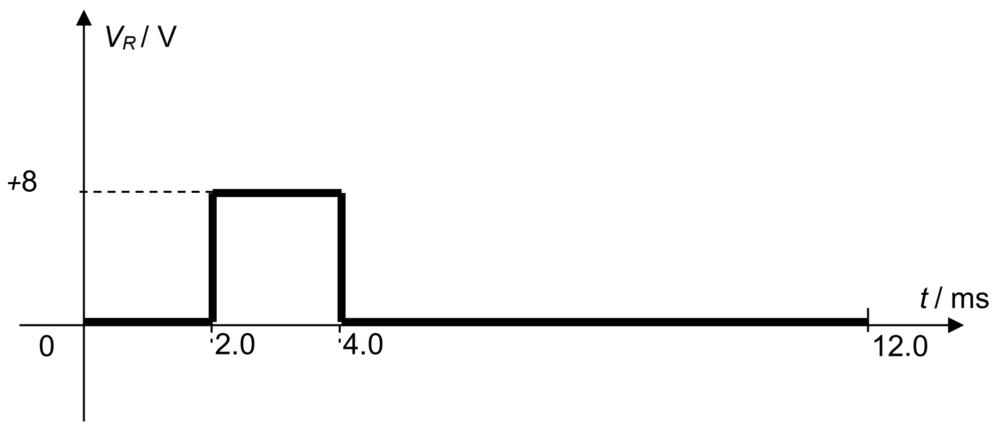
5	a	By transmitting electrical power in a.c. mode, the transmission voltage can be stepped up. This will reduce power loss in the cables due to a smaller transmission current.	M1 A1
	b(i)	$V_{rms} = \sqrt{\dfrac{2(10.0)^2(2.0)+(20.0)^2(2.0)+2(10.0)^2(2.0)+(20.0)^2(2.0)}{12.0}}$ $= 14\text{ V}$	C1 A1
	(ii)1.	0 V	A1
	(ii)2.	p.d. across resistor R = 20 – 12 =8.0 V	C1 A1
	(iii)	 correct shape of graph. value + 8 V labelled (awarded if graph is correct)	M1 A1
		Total:	9